

Rajalakshmi Engineering College, Thandalam 602105

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science and Engineering

CS23721 - Project Phase I

Review Form (Zeroth Review)

Team No: T076

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Project Title : Voice-Based AI Receptionist Agent for Smart College Communication.

Domain : Artificial Intelligence and Natural Language Processing.

SDG Goal No : 4, 9 and 16.

- SDG 04 - Quality Education.
- SDG 09 - Industry, Innovation and Infrastructure.
- SDG 16 - Peace, Justice and Strong Institutions.

Abstract

Educational institutions receive a large number of telephone inquiries every day from prospective students, parents, alumni, and existing students regarding admissions, academic programs, fee structure, hostel facilities, placements, examination schedules, and various student services. Most colleges continue to rely on manual reception systems, where staff members answer repetitive queries throughout the day. This often leads to long waiting times, inconsistent responses, increased workload, and limited service availability outside working hours. During peak periods such as admissions and examinations, handling a high volume of calls becomes increasingly difficult, affecting the overall quality of communication and user satisfaction. Furthermore, there is no intelligent mechanism to provide personalized information or seamlessly transfer complex queries to the appropriate personnel while maintaining conversation context.

This project proposes a **Voice-Based AI Receptionist Agent for Smart College Communication** that utilizes **Retrieval-Augmented Generation (RAG)**, **Large Language Models (LLMs)**, **Speech-to-Text (STT)**, **Text-to-Speech (TTS)**, and **Agentic AI** to automate institutional communication. The system answers telephone inquiries by retrieving information from a verified college knowledge base, ensuring accurate and context-aware responses. It

supports natural voice conversations, understands follow-up questions, and provides 24×7 assistance without human intervention for routine queries. For personalized student requests such as attendance, examination results, fee status, and placement eligibility, the system securely integrates with the college ERP after user authentication. When a query requires administrative judgment or falls outside the AI's confidence threshold, the agent intelligently transfers the call to a human receptionist while sharing the conversation summary for continuity. The proposed solution enhances communication efficiency, reduces receptionist workload, improves response consistency, and delivers an accessible, scalable, and intelligent information service that strengthens the overall digital transformation of educational institutions.

Comments / Remarks: (Filled by Panel Member)

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